MATH2241 MIDTERM TEST SOLUTION

(1) Since $x_n > 0$, we obtain $x_n = \frac{a_n + \sqrt{a_n^2+4}}{2}$ by quadratic formula. Suppose $(a_n) \to L \in \mathbb{R}$. Then Proposition 2.5 of the notes implies that

$$\lim_{n\to\infty} x_n = \frac{\lim_{n\to\infty} a_n + \sqrt{(\lim_{n\to\infty} a_n)^2 + 4}}{2} = \frac{L + \sqrt{L^2+4}}{2}.$$

(2) If (a_n) is not bounded, then one can find a monotone subseqence that converges to ∞ or $-\infty$ as follows. Suppose (a_n) is not bounded above (the other case is similar). Then for $C > 0$, there exists infinitely many $n \in \mathbb{N}$ such that $a_n \geq C$. We construct a subseqeunce inductively as follows. Define $a_{n_1} = a_1$. For $k \in \mathbb{N}$, the set $S_k := \{a_n : \ a_n \geq a_{n_k}\}$ is infinite. We can pick $a_{n_{k+1}} \in S_k$ such that $n_{k+1} > n_k$. Then we have an increasing subsequence.

If (a_n) is bounded, then there is a convergent subsequence by Bolzano-Weierstrass Theorem. Without loss of generality, we assume (a_n) converges to $L \in \mathbb{R}$ by taking subsequence. If there is infinitely many $a_n = L$, we are done. Otherwise, either there exist infinitely many $a_n > L$ or infinitely many $a_n < L$. We consider the first case since the other is similar. By taking a subsequence, we assume that

$$a_n > L \quad \forall n \in \mathbb{N} \qquad \text{and} \qquad (a_n) \to L.$$

We construct a subseqeunce inductively as follows. Define $a_{n_1} = a_1$. For $k \in \mathbb{N}$, the set $S_k := \{a_n : \ L < a_n < a_{n_k}\}$ is infinite. We can pick $a_{n_{k+1}} \in S_k$ such that $n_{k+1} > n_k$. We are done.

(3) Let $(a_n = \frac{n!}{n^n})$. Then $0 < a_{n+1}/a_n = \frac{1}{(1+1/n)^n} \leq \frac{1}{2}$ by binomial expansion. Hence, $\limsup |a_{n+1}/a_n| \leq 1/2 < 1$ and Ratio Test implies that (s_n) is convergent.

(4) Since $\lim_{x\to 0} f(x) = L$, given $\epsilon > 0$ there exists $\delta > 0$ such that $|f(x) - L| < \epsilon$ whenever x satisfies $0 < |x| < \delta$. Now whenever x satisfies $0 < |x| < \sqrt{\delta}$ (positive), we obtain $0 < |x^2| < \delta$ and thus $|f(x^2) - L| < \epsilon$. As $\epsilon > 0$ is arbitrary, we obtain $\lim_{x\to 0} f(x^2) = L$ by definition.